

REQUIREMENT ANALYSIS

Data Flow Diagram & User Stories

Date	01 JULY 2026
Team ID	
Project Name	Strategic Product Placement Analysis
Maximum Marks	4 Marks

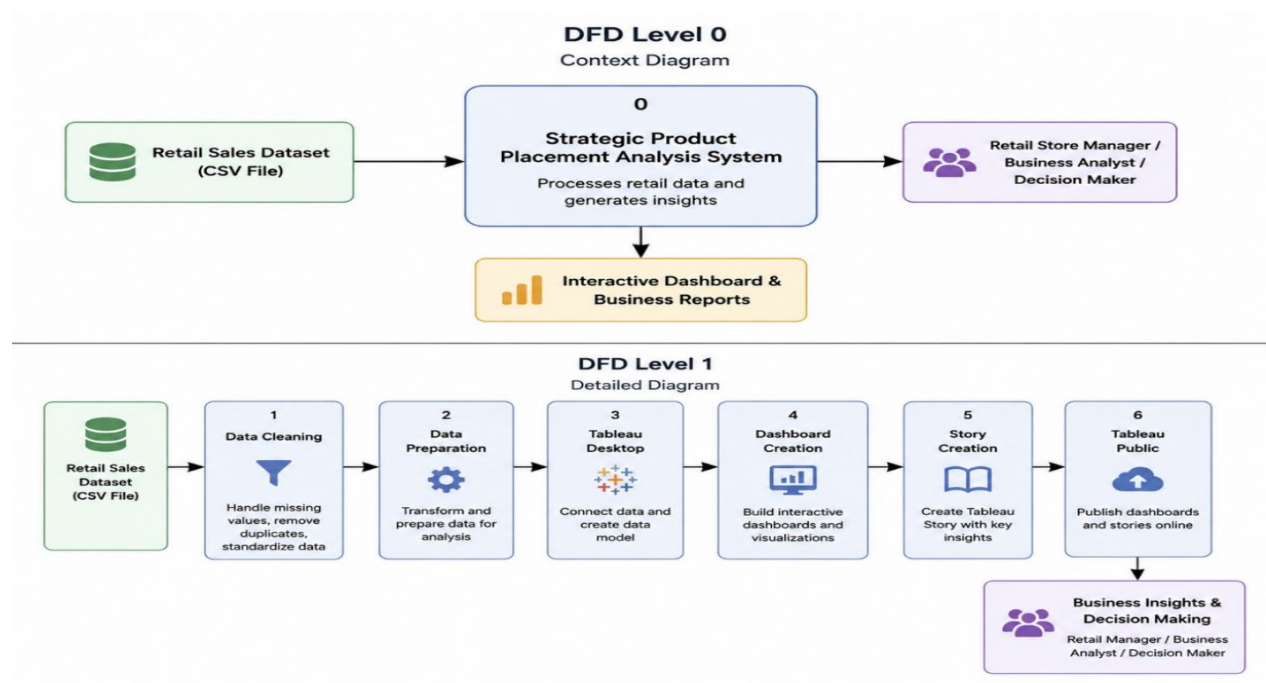
Data Flow Diagrams

A **Data Flow Diagram (DFD)** is a graphical representation of how data moves through a system. It illustrates the flow of information from the input source through various processing stages to the final output. DFDs help stakeholders understand how the system processes data, where data is stored, and how different users interact with the system.

For the **Strategic Product Placement Analysis** project, the DFD demonstrates how the retail sales dataset is collected, processed, analyzed using Tableau Desktop, and transformed into interactive dashboards and stories that support business decision-making.

The proposed solution consists of two levels of DFD:

- **DFD Level 0 (Context Diagram)** – Shows the interaction between users and the entire system.
- **DFD Level 1 (Process Breakdown)** – Shows the internal processes involved in transforming raw retail data into business insights.



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DFD Level 0 – Context Diagram

Description

The Retail Store Manager and Business Analyst provide the retail sales dataset to the Strategic Product Placement Analysis System. The system processes the data, generates interactive dashboards, Tableau Stories, and business reports, which are then used by decision-makers to improve product placement strategies and increase sales performance.

Components of DFD Level 0

Entity	Description
Retail Store Manager	Uploads dataset and reviews dashboard insights.
Business Analyst	Analyzes dashboard and prepares business reports.
Strategic Product Placement Analysis System	Processes retail sales data and generates visual analytics.
Tableau Dashboard	Displays interactive charts and KPIs.
Tableau Story	Presents business insights in sequence.
Business Reports	Supports retail decision-making.

DFD Level 1 – Process Breakdown

Description

The Level 1 DFD explains the internal working of the system. The retail sales dataset is first cleaned and validated before being imported into Tableau Desktop. Various analytical processes generate charts, KPI cards, dashboards, and stories. The completed dashboard is published on Tableau Public, while project documentation is maintained in GitHub.

Process Flow

Process ID	Process Name	Description
P1	Dataset Collection	Import Product_Positioning.csv dataset.
P2	Data Cleaning	Remove duplicates, handle missing values, validate records.
P3	Data Preparation	Organize fields and create calculated measures.
P4	Tableau Analysis	Generate charts, KPIs, and visualizations.
P5	Dashboard Development	Build interactive Tableau dashboard.
P6	Story Development	Create Tableau Story for business insights.
P7	Publishing	Publish dashboard on Tableau Public.
P8	Documentation	Upload reports and project files to GitHub.

Data Stores

Data Store	Purpose
D1 – Product_Positioning.csv	Stores retail sales data.
D2 – Tableau Workbook (.twbx)	Stores dashboard and story.
D3 – GitHub Repository	Stores documentation and project files.

External Entities

Entity	Interaction
Retail Store Manager	Uses dashboard for decision-making.
Business Analyst	Performs retail sales analysis.
Decision Maker	Reviews reports and recommendations.

Expected Outputs

Output	Purpose
Interactive Dashboard	Analyze product placement and sales.
Tableau Story	Explain insights sequentially.
KPI Dashboard	Monitor retail performance.
Business Reports	Support management decisions.
Tableau Public Dashboard	Online visualization.
GitHub Repository	Documentation and version control.

Advantages of the Proposed DFD

- Clearly represents data movement within the system.
 - Simplifies understanding of project workflow.
 - Supports better documentation.
 - Helps identify processing stages.
 - Improves communication among stakeholders.
 - Provides a clear overview of system architecture.
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User Stories

User Type	Epic	User Story	Priority
Retail Manager	KPI Dashboard	View sales KPIs	High
Retail Manager	Product Category	Compare category performance	High
Retail Manager	Promotion Analysis	Analyze promotional effectiveness	High
Business Analyst	Competitor Pricing	Compare prices with competitors	Medium
Business Analyst	Seasonal Analysis	Identify seasonal sales trends	Medium
Business Analyst	Customer Demographics	Analyze customer preferences	Medium
Decision Maker	Business Reports	Generate reports for management	High
Dashboard User	Interactive Filters	Filter dashboard easily	High